Logbook — Anomalous transport in soaring flights

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Working log

Working notebook. The timeline is regenerated from the git history; the narrative notes are written by hand.

1 Timeline (auto-generated from git)

Date	Event (commit subject)
2026-07-01	Fix delta config: season start 2001, data_root /Volumes/SSD_DISANTE/hang_gliders/delta_cfd_igc
2026-07-01	Add soaring-delta CLI for hang-glider CFD data (delta.ffvl.fr)
2026-07-01	Add analysis sub-package and preprocessing diagnostics; revise thesis chapters 1-4
2026-07-01	Expand pre-processing: explicit cleaning and filtering sections
2026-07-01	Clarify stationarity/compatibility check paragraph
2026-07-01	Add preliminary characterization section before main observables
2026-07-01	Add isotropy test protocol to the propagator item
2026-07-01	Rewrite propagator item: formal derivation of marginal scaling
2026-07-01	Reorganize TA-MSD item for clarity and typographic structure
2026-07-01	Clarify ergodicity breaking and add CTRW appendix
2026-06-28	Revise chapter 4 observables and the IGC description
2026-06-28	Warn when thesis and logbook next-steps drift apart
2026-06-28	Align logbook next steps with the thesis planning chapter
2026-06-28	Publish the logbook alongside the thesis
2026-06-28	Add living thesis document and build automation
2026-06-26	Deploy docs via GitHub Pages Actions artifact instead of gh-pages branch
2026-06-26	Fail fast with a clear message when data_root is unset
2026-06-26	Add 'verify' subcommand and remove machine-specific data_root path
2026-06-26	Link published docs in README; drop Development section
2026-06-25	Add GitHub Pages deployment via gh-pages branch
2026-06-25	Translate all documentation, comments, and strings to English
2026-06-25	Initialize thesis repo: FFVL CFD .igc data acquisition

2 Narrative notes

2026-06 — Paraglider data acquisition complete. The full FFVL paraglider CFD archive (`parapente.ffvl.fr`, seasons 1999–2000 to 2025–2026) has been scraped: about 203,000 flight records, of which ∼186,000 carry a GPS track. All 186,052 available tracks were downloaded. *What worked:* impersonating a Chrome TLS fingerprint (`curl_cffi`) reliably passes the FFVL Cloudflare challenge; the resumable, atomic, validated download survived many interruptions without corruption. *What did not / gotchas:* 173 tracks could not be obtained — 172 because the server returns something that is not a valid IGC (broken or placeholder links on FFVL's side, so a retry does not help), and 1 due to a transient Cloudflare challenge (worth a later retry). On macOS/exFAT, the OS scatters `._*` sidecar files next to every written file; a `clean` step removes them.

2026-06 — Integrity verified. A full on-disk re-scan (`verify`) confirmed that all 186,052 downloaded files are structurally valid IGC: no truncated files, no leftover `.part`, no read errors. The verification logic was promoted from a throwaway script to a first-class `soaring-para verify` subcommand (reusing the download-time validator), with tests.

2026-06 — Robustness of the data path. The external disk remounted under a different name (`HDD_DISANTE` → `SSD_DISANTE`), which would have silently broken a hard-coded path. Fix: removed every machine-specific path from the repo (placeholder + `SOARING_PARA_DATA_ROOT` environment variable) and added a start-up guard that aborts with a clear, actionable message when `data_root` is unset/unmounted — and verified that `/Volumes` is not user-writable, so a wrong path can never cause a silent download to the wrong place.

2026-07 — Hang-glider data acquired; naming corrected. The hang-glider CFD (`delta.ffvl.fr`, seasons 2001–2002 to 2025–2026) was added as a second source. The acquisition code (`soaring.acquisition.ffvl`) is shared; a second CLI entry point `soaring-delta` (config `delta_download.yaml`, env var `SOARING_DELTA_DATA_ROOT`) drives it. Data: ∼9,300 flights, 6,716 tracks downloaded, 30 failures (broken server-side links, same pattern as paragliders). Stored on the SSD at `/Volumes/SSD_DISANTE/hang_gliders/delta_cfd_igc/`. At the same time the paraglider CLI was renamed from the generic `soaring-ffvl` to `soaring-para` (and `SOARING_FFVL_DATA_ROOT` → `SOARING_PARA_DATA_ROOT`), and the config was renamed from `ffvl_download.yaml` to `para_download.yaml`, to avoid ambiguity now that both sources are present.

2026-06 — Documentation. Project docs are built with MkDocs and published on GitHub Pages. Initially deployed with `mkdocs gh-deploy`, which accumulated one built-site commit per deploy on a `gh-pages` branch; migrated to the official GitHub Pages *artifact* workflow (build → upload artifact → deploy), removing the branch and keeping the git history clean.

Operational note. In this repo `uv sync` occasionally fails to (re)install the editable package; `uv sync -reinstall-package soaring` fixes it.

3 Next steps

These mirror the planning chapter of the state document (`thesis/sections/04-next-steps`) and are kept in sync with it. The near-term plan is to analyse the data *in parallel* with studying the segmentation, since many observables can be built before any phase splitting.

- **Data sources.** FFVL paraglider and hang-glider CFD data are collected. Build the hang-glider catalog (`soaring-delta build-catalog`). Extend to further sources (XContest, possibly others), merging into one catalog and stratifying by source. Refresh `data/seasons_index.csv` via `soaring-para build-catalog` so the document statistics stay current.
- **Pre-processing.** Parse the IGC `B` records into clean, resampled, quality-filtered trajectories in a local metric frame.

- **Whole-trajectory observables (no segmentation).** Immediate next step: characterise the data with observables that need no phases — MSD and its scaling exponent, the displacement distribution and its tails, the velocity autocorrelation and turning-angle distribution, the speed/vertical-velocity distributions, and geometric measures (tortuosity, fractal dimension, radius of gyration). Goal: a first physical/statistical picture and some preliminary findings.
- **Segmentation.** Study Jérémie's internship report and code, then re-examine the transition/search/climb HMM (binary features: sign of vertical speed, curvature-angle persistence, straightness) — decide whether to keep that approach or add complexity for a better segmentation on our larger, more heterogeneous data.
- **Phase-resolved observables.** After segmentation, extract step lengths, waiting times, search durations, per-phase kinematics and the angular diffusion between successive glides.
- **Reproduce the reference.** Check that the enlarged dataset reproduces Vilpellet et al.: per-phase conditional MSD (their Fig. 3) and per-phase descriptive statistics (their Fig. 4), plus the global $H \approx 0.88$ — validation + universality test.
- **Transport analysis.** Estimate the MSD scaling exponent, fit the tails of the waiting-time and step-length distributions, and perform CTRW-vs-Lévy-walk model selection.
- **Minimal model.** A minimal stochastic model for $H \approx 0.88$: resolve transition segments, merge search+climb into one effective state, with angular diffusion between successive transition directions (preliminary idea, to be refined/changed).